Nathan Bellew

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Senior Software Engineer | Backend • Distributed Systems • Cloud • Security

Senior Software Engineer with 7+ years of experience designing and delivering production software across defense, cybersecurity, AI, and commercial technology environments. Strong background in Python, backend engineering, distributed systems, cloud infrastructure, containerization, and secure communications. Experienced owning systems from architecture and implementation through CI/CD, deployment, production troubleshooting, and long-term support. Proven technical leader with experience mentoring engineers, defining engineering standards, and translating complex product requirements into reliable production systems.

WORK EXPERIENCE

Integrity Communications Solutions, Inc. **Jun 2024 - Present**
Head of Software and Customer Experience • Colorado Springs, Colorado, United States

- Lead products from research and prototyping through production deployment, field support, maintenance, and iterative improvement.
- Design software architecture and engineering standards for reliability, scalability, maintainability, security, automated testing, and CI/CD.
- Lead and mentor cross-functional engineering personnel while establishing technical ownership, development practices, and execution priorities.

Principal Software Engineer & Senior AI Engineer • Colorado Springs, Colorado, United States

- Architected distributed production systems spanning backend services, edge devices, APIs, databases, cloud infrastructure, security, and automated deployment.
- Troubleshoot complex production failures across application software, Docker containers, Linux systems, networking, cloud infrastructure, and integrated hardware.
- Define software and AI technical strategy and align engineering architecture with product roadmaps and long-term business requirements.

Lockheed Martin **Jan 2020 - May 2024**
Software Engineer • Colorado, United States

- Maintained and enhanced satellite flight software and supporting simulation environments using Ada, C++, and Java.
- Served as Scrum Master within a SAFe Agile environment, coordinating team workloads, sprint execution, and release planning.
- Acted as Product Owner for software components, leading discussions around new features, defect resolution, prioritization, and release scope.

Cyber Systems Security Engineer Associate • Colorado Springs, Colorado Area

- Automated portions of the security assessment and reporting process, reducing manual effort required to organize findings and prepare system security documentation.
- Designed and developed a Python/Flask application that consolidated cybersecurity scan results and generated reports formatted for use in eMASS documentation and compliance activities.
- Built a lightweight SIEM capability using PowerShell to collect, normalize, and analyze security-related system data for cybersecurity monitoring and assessment workflows.

Colorado Mesa University **Feb 2019 - Jan 2020**
Team Lead for Cyber Security Student Center • Grand Junction, Colorado Area

- Led a 3–5 person cybersecurity development team and coordinated technical project execution.
- Led development of an AI-based malicious web-page and URL classification system.

EDUCATION

Masters of Science in Computer Science
Colorado Technical University • Colorado Springs, Colorado • GPA: 3.6 Jan 2021 - Expected Dec 2026

Bachelor's Degree in Major - Computer Science || Minor - Mathematics
Colorado Mesa University • GPA: 3.5 Jan 2015 - Dec 2019

CERTIFICATIONS

Active Top Secret Security Clearance | SCI Eligible

PROJECTS

BLDS Jun 2024 - Present

Integrity Communications Solutions

BLDS is an end-to-end radar-based monitoring platform I helped design, build, deploy, and maintain. The system combines mmWave radar hardware, Raspberry Pi edge controllers, Python-based processing, backend APIs, databases, containerized services, cloud infrastructure, and web interfaces to provide real-time occupancy, movement, and vital-sign monitoring without relying on wearable devices.

My work has included Python and object-oriented software development, radar and sensor integration, REST APIs, PostgreSQL and Redis, Docker, Linux, networking, CI/CD, secure device communication, JWT/TLS authentication, key management, automated deployment, fleet management, and production troubleshooting. I have also contributed to the broader system architecture, including device enrollment, server assignment, heartbeats, software updates, and centralized management through supporting services such as AWS Guide and the Auto Setup Service.

The project has progressed from early prototyping into a full production-oriented product, requiring development across embedded systems, backend services, infrastructure, security, deployment, testing, and long-term maintenance.

Deep Learning Malicious URLs Jun 2019 - Present

We are detecting and classifying malicious urls using Deep Learning Techniques. We trained ai to detect and classify urls as either benign, defacement, malware, phishing, or spam, using the deep learning frameworks Fast.AI and Keras-Tensorflow.

SKILLS

Agile, Automated Testing, AWS, Backend Development, Bash, C++, CI/CD, Confluence, Containerization, Database Design, DevOps, Distributed Systems, Docker, Docker Compose, Dockerfiles, Flask, Git, GitLab, GitLab CI/CD, Jira, Linux, Networking, PostgreSQL, Pytest, Python, Raspberry Pi, Redis, REST APIs, SAFe, Scrum, Software Architecture, SQL, TLS/JWT